

LAKSHMANA K

Boston, MA 02120

☎ (857)-397-5706 ✉ lakshmana.2003k@gmail.com [in linkedin.com/in/lakshmana-k-785727291](https://www.linkedin.com/in/lakshmana-k-785727291)

Education

Northeastern University — Khoury College of Computer Sciences

Sep. 2025 – May 2027

Master of Science in Computer Science (MSCS)

GPA: 3.665

VIT University

Sep. 2021 – Jul. 2025

B.Tech in Computer Science and Engineering with a Specialization in Internet of Things

GPA: 8.22

Technical Skills

Languages: Python, C++, C, Java, JavaScript, TypeScript, SQL, R

ML & Data: PyTorch, TensorFlow, NumPy, SciPy, Pandas, Scikit-learn

Frameworks & Tools: React, Next.js, Flask, Streamlit, MongoDB, Pinecone, Git, Docker, Pytest

Experience

Tech Mahindra

Jan 2025 – Jul 2025

Project Intern

Hyderabad, India

- Built a real-time monitoring dashboard decoding 10,000+ sensor packets per day, improving monitoring accuracy by 25%, using **Python**, **Flask**, and **MongoDB**.
- Developed an NLP-based internal project recommender over 300+ documents, reducing search time by 40%, using **Python**, **Scikit-learn**, and **Pinecone** for vector similarity search, delivered with a 6-member cross-functional team.

ThoughtGreen Technologies Pvt Ltd

Jun 2024 – Jul 2024

SDE Intern

Hyderabad, Telangana

- Contributed to the design and implementation of scalable software solutions alongside the development team, using **Python** and **Git**.
- Refined application features through user feedback sessions and participated in code reviews and debugging, improving code quality and user experience.

Hallmark Global Technologies, Inc.

Jun 2022 – Jul 2022

Web Developer Intern

Hyderabad, Telangana

- Designed and developed responsive web applications using **JavaScript**, **React**, and **HTML/CSS**, enhancing user engagement and functionality.
- Collaborated with cross-functional teams to define requirements and conducted performance testing to improve responsiveness.

Projects

Traffic Signal Control: Reinforcement Learning Comparison | *Python, PyTorch, NumPy, Matplotlib*

- Built a custom traffic-intersection simulator and implemented six control policies — DQN, Double DQN, Dueling DQN, tabular Q-Learning, SARSA, and a fixed-time baseline — in a shared training and evaluation harness (team project, 2 authors).
- Diagnosed why initial learned agents underperformed the baseline (reward misalignment and excessive phase-switching) and redesigned the reward around total queue length, adding a minimum-green constraint and a phase-duration state feature.
- After the redesign, the deep-RL agents reduced average vehicle waiting time by roughly 78% versus the fixed-time baseline while maintaining throughput, evaluated over 20 episodes.

Smart Water Management System | *IoT, Embedded C, Mobile App*

- Built an IoT system with sensors to monitor water levels, usage, and quality in real-time, optimizing water resource management.
- Implemented automated control and remote monitoring via a connected mobile app, minimizing wastage and enabling proactive maintenance through timely alerts.

Food Quality Monitoring-IoT | *IoT, Embedded C, Mobile App*

- Designed and deployed an IoT system to monitor food quality parameters like temperature and humidity, enhancing monitoring accuracy.
- Created a notification system to alert users via mobile applications when food quality parameters deviate from predefined thresholds.

Certifications

GCCF Certified